

04c — SOAP

Key Concepts

Term	Definition
SOAP	Simple Object Access Protocol — strict XML-based protocol for APIs
WSDL	Web Services Description Language — mandatory XML contract describing the service
Envelope	Wrapper element of every SOAP message
Fault	SOAP's built-in error structure (inside Body)
WS-Security	Built-in SOAP standard for message-level security

SOAP vs REST

	SOAP	REST
Type	Protocol — strict rules	Architectural style — flexible
Data format	XML only	JSON, XML, anything
Transport	HTTP, SMTP, TCP	HTTP only
Contract	WSDL (mandatory)	OpenAPI (optional)
Error handling	Built-in Fault structure	HTTP status codes
Speed	Slower — verbose XML	Faster — lightweight JSON
Use case	Enterprise, banking, legacy	Web, mobile, modern APIs

SOAP Message Structure

```
<soap:Envelope>
  <soap:Header>
    <!-- auth tokens, metadata, routing -->
  </soap:Header>
  <soap:Body>
    <GetUserRequest>
      <UserId>1</UserId>
    </GetUserRequest>
  </soap:Body>
</soap:Envelope>
```

Part	Purpose
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Part	Purpose
Envelope	Marks it as a SOAP message
Header	Optional — auth, routing, transaction info
Body	Actual request or response data
Fault	Error details when something fails

Why SOAP Still Exists

- **Strict contracts** — WSDL guarantees both sides agree on exact structure
- **Built-in standards** — WS-Security, WS-ReliableMessaging, WS-Transaction
- **Legacy systems** — rewriting 20-year-old banking infrastructure is too risky and expensive
- **Audit trails** — XML messages are easy to log and verify for compliance

Industry Examples

Company/System	Usage
PayPal	Older APIs were SOAP, now REST but SOAP still supported
Salesforce	Still offers SOAP API alongside REST
Banking/SWIFT	Interbank communication uses SOAP-based services
Government APIs	Older integrations (e.g. GST) are SOAP-based

Interview Questions

Q: What is SOAP and how does it differ from REST? A: SOAP is a strict protocol using XML only and requiring a WSDL contract. REST is a flexible style using JSON over HTTP. SOAP is heavier but has built-in security and transaction standards.

Q: What is WSDL? A: Web Services Description Language — an XML file that acts as a mandatory contract, describing what operations the service offers, what parameters they take, and where the service lives.

Q: Why do enterprises still use SOAP? A: Built-in WS-Security and transaction standards, strict contracts for compliance, and legacy systems too expensive to rewrite. Not preference — cost of change.